ALICE L WALTON SCHOOL OF MEDICINE

Nonopioid Analgesics – Aspirin

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Adapted from: Antiplatelet Drugs

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0:00 / 15:26 1x

Learning Objectives (4) Versions

After completing this brick, you will be able to:

1. Explain the mechanism of action and therapeutic uses of aspirin.
2. Justify the use of low doses of aspirin for cardiovascular protection, as opposed to higher doses for pain and inflammation, by explaining the mechanism through which aspirin affects platelet aggregation.
3. Identify potential side effects associated with aspirin and explain underlying pathophysiologic mechanism.
4. Apply the knowledge of aspirin's pharmacology to provide rationale for evidence-based recommendations in various clinical scenarios.

CASE CONNECTION

EG is a 6-year-old girl who goes to kindergarten. EG comes home from kindergarten and complains of fever, body ache, and headache. When her mother asks her to eat, she refuses to have any food. Her mother calls the school nurse to find out if she was okay in school. The school nurse states that a few of the students in EG's class have a viral infection and have similar symptoms. EG's mother has Aspirin in her medicine cabinet.

Should EG's mother give her Aspirin for her fever and body ache? Should EG's mother completely avoid giving her Aspirin?

Consider your answers as you read, and we'll revisit EG at the end of the brick.

GO TO CONCLUSION ↓

Cyclooxygenase Inhibitors

Aspirin (Acetylsalicylic acid) belongs to the NSAID class of drugs. They are one of most commonly used agents that are recommended for a wide variety of uses. You may recall the figure below from the NSAIDs brick. Note the arm that leads to COX-mediated byproducts, in particular Thromboxane-A2. (Figure 1)

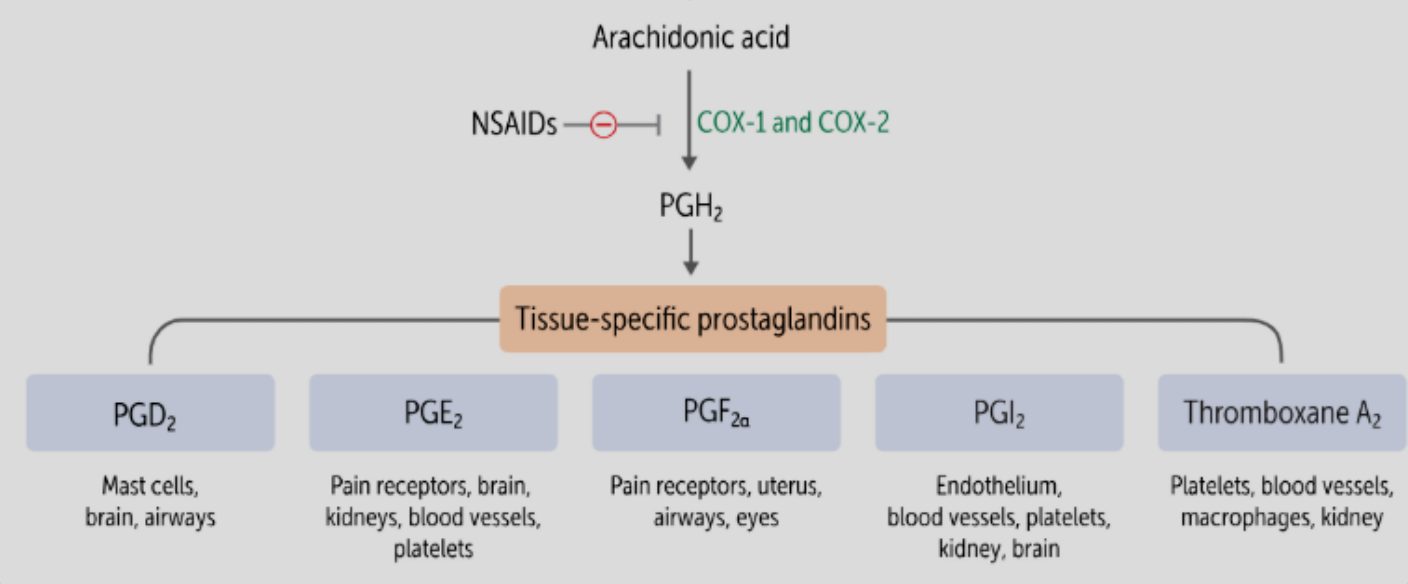
Diverse physical, chemical, inflammatory, and mitogenic stimuli
Phospholipase A₂
Arachidonic acid
NSAIDs
COX-1 and COX-2
PGH₂
Tissue-specific prostaglandins
PGD₂
Mast cells, brain, airways
PGE₂
Pain receptors, brain, kidneys, blood vessels, platelets
PGF₂α
Pain receptors, uterus, airways, eyes
PGI₂
Endothelium, blood vessels, platelets, kidney, brain
Thromboxane A₂
Platelets, blood vessels, macrophages, kidney

Figure 1

How does Aspirin work? (LO#1)

As stated in the NSAID brick, Aspirin was classified as a non-selective COX inhibitor and as a salicylate. Chemically, Aspirin is acetylsalicylate. At higher doses, ASA irreversibly inhibits COX-2, preventing the production of prostacyclin. Higher doses of aspirin (>75mg) inhibit PG synthesis, and thus, aspirin can be used as an analgesic and antipyretic. These actions are similar to those of the NSAIDs. Additionally, Aspirin can also inhibit platelet aggregation by the inhibition of thromboxane- A2 (TxA2) and is used as an anti-platelet agent. Once acetylated, COX-1 is completely inactivated, whereas acetylation of COX-2 converts enzyme activity from a cyclooxygenase to a lipoxygenase (LOX).

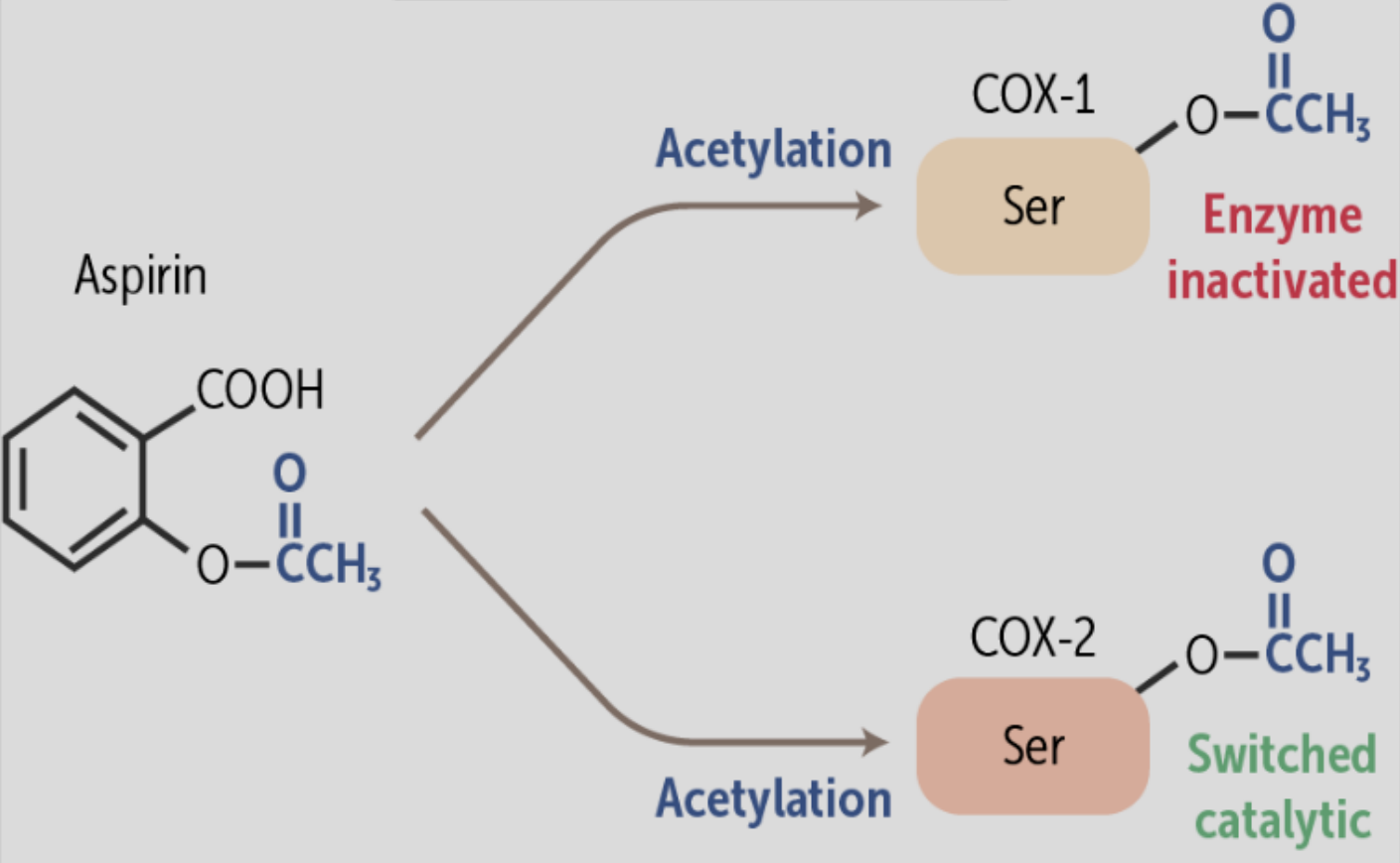

Figure 2

How does aspirin inhibit platelet aggregation?

How does Aspirin help with lowering a cardiovascular event? (LO#2, 4)

Platelet aggregation, among other factors, is a cause of myocardial infarctions. TxA2 is important for platelet aggregation. Aspirin irreversibly inhibits TxA_2, thus preventing platelet aggregation. Since platelets are non-nucleated and do not contain DNA to support transcription and translation, inhibition of the COX enzyme lasts for the lifespan of the platelets (~7-10 days). If the drug is not continued, the platelet inhibitory function diminishes. In patients with preexisting vascular disease, the use of low or higher doses of Aspirin was equally effective in preventing a cardiovascular event. Aspirin in doses from 75 to 325 mg daily produces similar benefits on cardiovascular events to high doses such as 1300 mg. However, the lower doses produce the therapeutic benefit with far fewer side effects. The use of low dose 81mg of Aspirin in patients with an existing cardiovascular event is very common.

CLINICAL CORRELATION

In acute cardiovascular events such as acute coronary syndrome (STEMI) or stroke, regular aspirin should be used to achieve a rapid clinical antithrombotic effect. If the only available preparation is enteric coated, the single tablet or multiple tablets necessary to achieve the recommended dose of 325 mg should be crushed or patients are asked to chew the tablet. This allows for rapid absorption of aspirin and its an[illegible]

What is the mechanism of action of ASA?

What are some adverse effects associated with Aspirin use? (LO#3, 4)

Pharmacokinetics and Pharmacodynamics of Aspirin

To understand, Aspirin's ADRs it is important to first understand its PK/PD. Aspirin has a pH of 3.5, and hence, it is mostly absorbed in the stomach (Pharmocokinetics1: 3of 3 pH and pKa)Recall the brick on pH and pKa). Aspirin is hydrolyzed to salicylic acid and is cleared renally. An increase in urinary pH can increase the clearance of Aspirin.

- ***GI-related effects:*** The most common ADRs related to Aspirin therapy are mainly in the sto[illegible]lcer, erosive gastritis, and upper gastrointestinal bleeding. GI-related ADRs can occur within minutes to months of use and occur due to the damage to the gastric mucosa caused by inhibition of PG function. This mechanism is similar to NSAIDs (discussed in NSAID brick). Thus, it is best to avoid aspirin in patients with active peptic ulcer disease (increase in bleeding risk). Aspirin's hydrolysis to salicylates follows first-order kinetics. Subsequently, the salicylates undergo phase II reactions (glucuronidation, oxidation) to be renally excreted. In case of an aspirin overdose, the hepatic enzymes are saturated, salicylate metabolism shifts to zero-order kinetics, and elimination decreases. This could lead to salicylate toxicity with metabolic acidosis.
- ***Hypersensitivity reactions:*** Hypersensitivity reactions of the skin (Urticaria, angioedema) and airways (dyspnea) can occur. Aspirin-exacerbated respiratory disease (AERD) involves respiratory distress and is a non-immunological reaction due to COX-1 inhibition. The increase in pro-inflammatory leukotrienes is one of the major causes of AERD. AERD can cause bronchoconstriction, vascular leaks, mucus secretion, and increased eosinophilic and mast cell-mediated inflammation. Aspirin is contraindicated in patients with Asthma, rhinitis, and nasal polyps. Hypersensitivity to NSAIDs is also a contraindication for Aspirin use.
- ***Reye's Syndrome:*** Aspirin should not be used in febrile children because of a concern for Reye's syndrome, which is rare but potentially fatal. This can cause rapid fatty liver degeneration leading to hepatic encephalopathy and potentially irreversible brain damage. Due to the possibility of Reye's syndrome, Aspirin, and other salicylates are contraindicated in infants, children, and children recovering and during viral infections (< 18 years)
- ***Tinnitus:*** At high doses, ASA has effects on the cochlea that can result in tinnitus.

CLINICAL CORRELATION

AERD presents during late childhood to adulthood, with the median age of onset around 34 years. Nasal congestion and rhinorrhea are usually the first symptoms to develop. Common sinonasal symptoms in AERD patients include nasal congestion or obstruction, anosmia, and postnasal drip. Patients with AERD often have rapid nasal polyp recurrence. Asthma can also develop on average 2 years after initial upper respiratory tract symptoms, and aspirin-intolerance can also occur (usually 4years later).

What are the absolute contraindications for giving ASA?

CASE CONNECTION

BACK TO INTRODUCTION ↑

Thinking back to EG, should EG's mother give her Aspirin for her fever and body ache? Should EG's mother completely avoid giving her Aspirin?

Aspirin is indicated for its analgesic and antipyretic properties. Like NSAIDs, it can inhibit COX and downstream production of prostaglandins to elicit these actions. However, note EG's age. She is 6 years old, and the use of Aspirin is contraindicated in infants and children. Moreover, the presence of a suspect viral infection also supports not giving the medication to EG. In this case, use of pediatric doses of NSAIDs (OTC or prescribed) or acetaminophen (OTC or prescribed) would be appropriate.

Summary

- Aspirin, a non-selective COX inhibitor, irreversibly inhibits COX-2 at higher doses, preventing the production of prostacyclin and inhibiting platelet aggregation.
- Aspirin's anti-platelet action can help lower the risk of cardiovascular events by inhibiting thromboxane A2 and, subsequently, platelet aggregation, a cause of myocardial infarctions.
- Low or higher doses of Aspirin are equally effective in preventing a cardiovascular event, with lower doses producing fewer side effects.
- Adverse effects of Aspirin use include GI-related effects, hypersensitivity reactions, Reye's Syndrome in febrile children, and tinnitus at high doses.
- Aspirin is used in the management of acute coronary syndrome and should be given as the first step in treatment for suspected myocardial infarction.
- Aspirin is contrai[illegible] ASA-related anaphylactic reaction, bleeding disorders, severe liver or renal failure, active or recent gastrointestinal or intracranial bleeding, or to febrile children.

Review Questions

1. Which enzyme does aspirin irreversibly inhibit at higher doses?

A. ○ COX-1
B. ● COX-2
C. ○ LOX
D. ○ PGE2

Hide Explanation

The correct answer is COX-2 (B). At higher doses, aspirin irreversibly inhibits COX-2, preventing the production of prostacyclin. COX-1 (A) is irreversibly inactivated by acetylation but not specifically at higher doses. LOX (C) is not directly inhibited by aspirin; rather, COX-2 activity is converted to LOX activity upon acetylation. PGE2 (D) is a prostaglandin, not an enzyme.

2. What is the primary mechanism by which aspirin prevents platelet aggregation?

A. ● TxA2 inhibition
B. ○ Reversible COX-2 inhibition
C. ○ Reversible COX-1 inhibition
D. ○ LOX inhibition

Hide Explanation

The correct answer is TxA2 inhibition (A). Aspirin irreversibly inhibits thromboxane A2 (TxA2), which is crucial for platelet aggregation. Reversible COX-2 inhibition (B) is incorrect because COX-2 inhibition by aspirin is irreversible. Reversible COX-1 inhibition (C) is incorrect as aspirin irreversibly inhibits COX-1. Lipoxygenase activation (D) is incorrect as it is not the primary mechanism for preventing platelet aggregation.

3. Which condition is a contraindication for Aspirin use?

A. ● Active treatment for gastric ulcer
B. ○ Tinnitus
C. ○ Smoking
D. ○ Concomitant NSAID use

Hide Explanation

The correct answer is active peptic ulcer disease (A). Aspirin can increase the risk of bleeding in patients with active peptic ulcer disease. Tinnitus (B) is a side effect of high doses, not a contraindication. Smoking (C) is not a contraindication. Concomitant NSAID use is not a contraindication (D). Hypersensitivity to NSAIDs is a contraindication is a potential outcome of Reye's syndrome, not a direct contraindication.

4. Aspirin is contraindicated in infants and children because they can develop ____.

A. ○ Dyspepsia
B. ● Reye's Syndrome
C. ○ Nausea
D. ○ Tinnitus

Hide Explanation

The correct answer is Reye's Syndrome (B). Reye's Syndrome is a rare but potentially fatal condition that can occur in children using Aspirin. Dyspepsia (A) can occur as an adverse effect, but it is not a contraindication. Nausea (C) can occur but is not irreversible. Tinnitus [illegible]gh doses.

irreversible. Tinnitus (D) is an adverse effect that can occur at high doses.

How would you rate this Brick?

Review Learning Objectives

Check off the objectives when you feel comfortable with the concept

01 ☑ ~~Explain the mechanism of action and therapeutic uses of aspirin.~~
02 ☑ ~~Justify the use of low doses of aspirin for cardiovascular protection, as opposed to higher doses for pain and inflammation, by explaining the mechanism through which aspirin affects platelet aggregation.~~
03 ☑ ~~Identify potential side effects associated with aspirin and explain underlying pathophysiologic mechanism.~~
04 ☑ ~~Apply the knowledge of aspirin's pharmacology to provide rationale for evidence-based recommendations in various clinical scenarios.~~

Objective Confidence: 4/4